# Factory Resume

Half the population disappears overnight. How do essentials still reach the people who are left?

**The idea in 30 seconds**

When half the people vanish, half the warehouse staff and half the customers vanish too. So the problem isn't "not enough capacity". It's that **the wrong people are left, people need different things, and help arrives late**. Factory Resume is a tool for the warehouse operations manager. Every morning it builds a plan: who works where, which partners to call, and which orders ship first. Each evening it learns from what actually happened.

**Result in the prototype (5 days):** 100% of essential items delivered on time vs. 59% without the tool, and 37% more orders shipped overall, for Rs 14 of partner cost per extra item.

## The situation the prototype models

- A personal-care and hygiene brand (fictional) with warehouses in Bengaluru and Pune. **60 of its 120 staff are left**, and couriers are down by half.
- **The wrong people are left:** the packing team lost the most people. Orders can only go out as fast as the slowest step, so packing now holds everything up.
- **Needs have changed:** people are stocking up on soap, hand wash and sanitary pads, and have stopped buying face serum. Shipping orders in date order would still send lip balm before soap.
- **Help is slow:** outside partners need 6 to 48 hours to start.

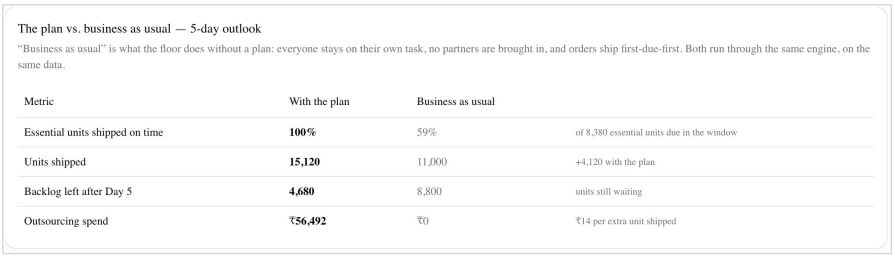
## Walkthrough: one day with Factory Resume

### Step 1

**See the whole picture (Overview)**

The manager first sees what the plan achieves against "business as usual": everyone stays in their usual job, no partners, and orders ship by due date.

Both are calculated by the same engine on the same data, so the comparison is measured, not claimed.



The plan vs. business as usual — 5-day outlook

"Business as usual" is what the floor does without a plan: everyone stays on their own task, no partners are brought in, and orders ship first-due-first. Both run through the same engine, on the same data.

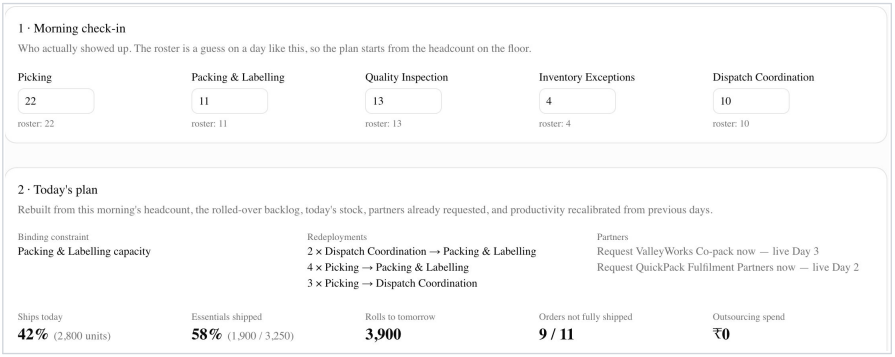
| Metric | With the plan | Business as usual | |
|---|---|---|---|
| Essential units shipped on time | **100%** | 59% | of 8,380 essential units due in the window |
| Units shipped | **15,120** | 11,000 | +4,120 with the plan |
| Backlog left after Day 5 | **4,680** | 8,800 | units still waiting |
| Outsourcing spend | **₹56,492** | ₹0 | ₹14 per extra unit shipped |

### Step 2

**Check in and get today's plan (Daily loop)**

At 7am the manager enters who actually turned up. The plan rebuilds instantly from that headcount, not from the roster.

The plan names the bottleneck (packing), the people to move, and the partners to call now so they're ready tomorrow. Every order shows why it's ranked where it is, e.g. "Essential · due in 2d".



1 · Morning check-in

Who actually showed up. The roster is a guess on a day like this, so the plan starts from the headcount on the floor.

| Picking | Packing & Labelling | Quality Inspection | Inventory Exceptions | Dispatch Coordination |
|---|---|---|---|---|
| 22 | 11 | 13 | 4 | 10 |
| roster: 22 | roster: 11 | roster: 13 | roster: 4 | roster: 10 |

2 · Today's plan

Rebuilt from this morning's headcount, the rolled-over backlog, today's stock, partners already requested, and productivity recalibrated from previous days.

Binding constraint: Packing & Labelling capacity

Redeployments:
2 × Dispatch Coordination → Packing & Labelling
4 × Picking → Packing & Labelling
3 × Picking → Dispatch Coordination

Partners:
Request ValleyWorks Co-pack now — live Day 3
Request QuickPack Fulfilment Partners now — live Day 2

| Ships today | Essentials shipped | Rolls to tomorrow | Orders not fully shipped | Outsourcing spend |
|---|---|---|---|---|
| **42%** (2,800 units) | **58%** (1,900 / 3,250) | **3,900** | **9 / 11** | **₹0** |

## Step 3

**Move people to where they're needed (Workforce)**

Some workers can cover a second job: a picker can pack at 60% speed after 2 hours of training. The tool moves one person at a time to the bottleneck, and only when that actually speeds up the line.

Day 1: 9 of 60 people are moved, so **2,800 items ship instead of 2,200**. Quality checkers are never moved, and no one untrained is moved into quality checks.

Redeployments in today's plan

A line ships only as fast as its slowest step. The engine moves one cross-skilled worker at a time from the best-covered task to the worst-covered one, and only when that lifts the weaker of the two. Moved workers are slower and lose ramp-up hours, and both costs are counted. Quality Inspection and Inventory Exceptions need certification, so no one is moved into them and their own staff are never moved out.

| Move | Workers | Speed on new task | Ramp-up | Units/shift given up | Units/shift gained |
|---|---|---|---|---|---|
| Dispatch Coordination → Packing & Labelling | 2 | 50% | 2h | −640 | +240 |
| Picking → Packing & Labelling | 4 | 60% | 2h | −800 | +360 |
| Picking → Dispatch Coordination | 3 | 50% | 1h | −600 | +263 |

## Step 4 · Call partners at the right time (Outsourcing)

Partners only count for the hours they're actually working. A partner that needs 24 hours to start can't help today, so the tool says “request it now for tomorrow”. It also won't pay for extra picking while packing is the bottleneck, because those items would just pile up.

## Step 5 · Ship what matters first (Recovery plan)

Each product is tagged essential, everyday or non-essential. Essentials go first, then whatever is about to miss its date. On Day 1 the plan ships 1,900 essential items; shipping by due date alone would ship none. The manager approves the plan, and nothing is booked automatically.

## Step 6

**Close the day, learn, and look ahead**

At the end of the shift the manager enters what each team actually did. If packing hit only 80% of plan, tomorrow's plan assumes a slower pace. Plan accuracy is tracked every day.

**Key insight:** the bottleneck moves every day: packing, then picking, then Pune's couriers, then stock. A plan made once would be fixing the wrong problem by Day 3. The Scenarios page tests what happens if things get worse.

The bottleneck moves — day by day

The plan if each day executes as planned. The constraint that binds changes as partners come online and stock runs down, which is why the plan is rebuilt every morning rather than once.

| Day | Ships | Essentials shipped | Partners live | Binding constraint | Backlog at close |
|---|---|---|---|---|---|
| Day 1 · 25 Oct | 2,800 | 1,900 | None | Packing & Labelling capacity | 6,520 |
| Day 2 · 26 Oct | 3,500 | 2,560 | 3 | Picking capacity | 5,640 |
| Day 3 · 27 Oct | 3,600 | 2,210 | 3 | Last mile — Pune FC | 4,660 |
| Day 4 · 28 Oct | 3,170 | 1,710 | 3 | Last mile — Pune FC | 4,110 |
| Day 5 · 29 Oct | 2,050 | 1,190 | 3 | Stock — FW-BODYLOTION-200 | 4,680 |

## Why it's built this way

- **Simple rules over a black box.** A manager has to defend every call to leadership, so every move the tool makes can be explained in one sentence.
- **A person stays in charge.** The tool recommends; the manager approves.
- **Uses data teams already have:** the order list, the staff roster, and stock levels. The riskiest assumption is that rosters record who can do which second job. I'd test that first with real managers.

## How success is measured

**Main goal:** essential items delivered by their promised date. **Trust:** the share of plans the manager approves without changes, and plan accuracy (shipped vs. planned). **Cost:** partner spend per extra item shipped. **Safety checks:** no rise in quality issues from moved staff, and customers told about delays before their date.